Inyoung Oh

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Korea Institute of Science and Technology (KIST), Seoul, Republic of Korea

Fusing semantic and geometric cues for reliable, metric 3D spatial perception.

Research Focus

My research makes learned and semantic 3D *metric, consistent, and structured* by fusing it with explicit geometry, recovering reliable spatial structure exactly where data-driven models are least faithful: at object boundaries, sharp discontinuities, and absolute scale.

- Scene-intrinsic geometric cues (surface normals, sharp boundaries, ground planes) as transferable structure injected into learned and semantic 3D
- Metric, consistent 3D from a single image: bridging 2D and 3D through scene geometry, where learned depth and multi-view reconstruction fall short
- Geometric representations that generalize across sensors and domains (CAD, LiDAR, indoor/outdoor scans, monocular)

Education

Ph.D., Mechanical and Robotics Engineering, Gwangju Institute of Science and Technology (GIST) **Feb 2026**

Dissertation: *Normal Vector Estimation, and Semantic Segmentation of 3D Point Clouds using Deep Learning and Geometric Analysis*

M.S., Mechatronics, GIST **Feb 2018**

Thesis: *Sphere and Cylinder Detection in Kinect Point Clouds using RANSAC and the 2D Hough Transform*

B.S., Mechatronics, Chungnam National University **Feb 2016**

Research Experience

Postdoctoral Fellow, Visual Intelligence Group, KIST | Advisor: Dr. Junghyun Cho **Feb 2026–Present**

- Reframed relocalization from a moving camera (placing it and nearby people in metric 3D from RGB alone) as a **geometric 2D–3D bridge**: on an up-to-scale map, metric placement reduces to a ray–ground intersection set by one geometric quantity, the ground-plane normal, independent of pose and reconstruction method.
- Bypassing monocular depth's principled limits, this estimator recovers metric position **7.7–100**× better than strong monocular-depth and recent VGGT-style feed-forward baselines (even given GT pose) across indoor and outdoor capture, isolating capture geometry, not any model or dataset, as the cause (manuscript in preparation [P1]).
- This puts my doctoral geometry at the crux: estimating scene-intrinsic structure (surface normals, sharp and planar discontinuities, robust to sparsity and noise) is exactly what my descriptors deliver, carried from point clouds into image-grounded 3D.

Research Assistant, Modeling and Simulation Lab, GIST | Advisor: Prof. Kwang Hee Ko **Feb 2016–Feb 2026**

Established the geometric half of this fusion: explicit scene-intrinsic cues (surface normals, sharp features), injected into learned models, that transfer across sensors, sampling density, and noise.

Sharp Feature Detection & Geometric Descriptors

- **SFD-Net** (ECCV 2026): an architecture-agnostic descriptor that detects sharp features as surface-normal discontinuities. Prepended to three independent backbones *without modification*, it raises Average F1 by +4–7 pp (the descriptor, not backbone-specific tuning, drives the gains) and transfers *zero-shot* from 10K-point synthetic CAD to real-world scans more than an order of magnitude larger (∼36×); SOTA Average F1 on ABC across all four noise conditions. [C1]
- Grounding the descriptor theoretically via directional statistics, replacing heuristic scale selection with a training-free geometric prior that stays informative precisely at discontinuities (in preparation). [P2]

Normal Vector Estimation

- A multi-scale normal estimator resolving neighborhood-scale ambiguity under non-uniform sampling via attention-based scale weighting and Chamfer-distance noise-aware supervision; SOTA on the PCPNet primitive subset (noise-free RMSE 1.821°, 6.53M parameters). [J1]

Semantic Segmentation (Outdoor & Indoor)

- Injected intensity-assisted normal cues into RandLA-Net via a Normal Local Feature Aggregation module, yielding +4.0 pp mIoU on outdoor LiDAR (SemanticKITTI, 53.9→57.9) with pronounced gains on safety-critical classes (Bicycle +21.4 pp, Motorcycle +15.2 pp, Pedestrian +4.0 pp). [J3]
- Extended refined normals and multi-scale geometric features into boundary-aware indoor segmentation on S3DIS, achieving +6.3% mIoU over the PushBoundary baseline (Ph.D. thesis).

6-DoF Pose Estimation & MR Systems

- Designed RoI-PCA color-space augmentation to decouple object appearance from background variation, improving 5cm5deg accuracy by +10.77pp on LINEMOD; integrated into a real-time, markerless multi-device MR remote collaboration system (MR glasses + projector–camera), without depth sensors. [J2]

Geometric Primitive Recognition

- Proposed a parameter-free pipeline-recognition method from unorganized point clouds (curvature-driven classification + sphere-RANSAC centerline recovery), recovering topology-consistent structure from raw scans without user-specified priors for radius, orientation, or connectivity. [J4]

Publications

Published / Accepted

[C1] **Inyoung Oh**; K.H. Ko. "SFD-Net: Sharp Feature Detection Network Based on Local Geometric Features." *European Conference on Computer Vision (ECCV)*, **2026**. *(accepted)*

[J1] **Inyoung Oh**; J. Song; M. Kim; D. Yun; K.H. Ko. "Attention-Guided Multi-Scale Point Cloud Normal Estimation with Noise-Aware Training." *Int. J. Image and Graphics*, **2026**. *(accepted)*

[J2] **Inyoung Oh**; G. Jang; J. Song; M. Son; D. Kim; J. Yun; K.H. Ko. "A Mixed Reality-based Remote Collaboration Framework Using Improved Pose Estimation." *Computers in Industry*, 174, 104414, **2026**.

[J3] M. Kim; **Inyoung Oh**; D. Yun; K.H. Ko. "Improved Semantic Segmentation Network Using Normal Vector Guidance for LiDAR Point Clouds." *J. Computational Design and Engineering*, 10(6), 2332–2344, **2023**.

[J4] **Inyoung Oh**; K.H. Ko. "Automated Recognition of 3D Pipelines from Point Clouds." *The Visual Computer*, 37(6), 1385–1400, **2021**.

In Preparation

[P1] **Inyoung Oh** et al. "Metric Relocalization in Up-to-Scale Maps via Ground-Plane Geometry."

[P2] **Inyoung Oh** et al. "A Directional Statistics Approach to Geometric Sharpness Estimation in 3D Point Clouds."

Patent

K.H. Ko; **Inyoung Oh**. "Automated System for Detecting 3D Pipeline." KR 2010-2135828 B1 (Registered), **July 2020**.

Awards & Scholarships

Best Poster Award (×6), Korean Society for Computational Design and Engineering (CDE)	**2020–2025**
CDE DX Encouragement Award, Korean CDE Society	**2022**
Outstanding PhD Student RA Scholarship (×5), GIST	**2018–2025**
GIST Full Scholarship (Government-funded, Doctoral & Master's)	**2016–2026**

Presentations

Inyoung Oh; K.H. Ko. "Automatic Detection of Cylindrical Objects from Unorganized Point Cloud." *EuroVR* (poster), London, UK. **2018**

Inyoung Oh; K.H. Ko. "A RANSAC-based Method for Detection of Multiple Spheres from a Point Cloud." *Computer Graphics International* (poster), Yokohama, Japan. **2017**

Teaching & Mentorship

Mentorship, Modeling and Simulation Lab, GIST: mentored M. Kim from undergraduate intern through graduate studies; mentee published a first-author journal paper (*JCDE*, 2023). Also guided additional interns and supervised a bachelor's thesis. **2018–2026**

Teaching Assistant, Engineering Analysis, GIST: recitations, assessment design, Python/MATLAB/C++ instruction. **2017, 2020**

Professional Service

Reviewer, *Journal of Computational Design and Engineering* (JCDE)

Technical Skills

Languages & Frameworks: Python, PyTorch, C++, MATLAB, Open3D, PCL, Unity3D
Sensors & Devices: Velodyne/Ouster LiDAR, Kinect v2, Azure Kinect, MR smart glasses

References

Prof. Kwang Hee Ko	**Prof. Bochang Moon**	**Prof. Jinho Song**
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